

Experiment 1:

1. For input=1:

Output
<<-----Menu----->> 1. Length 2. Copy String in another String 3. Reverse String 4. Check for Palindrome 5. Check for Substring 6. Compare Any String with another 7. Display Menu 8. Exit ENTER CHOICE (7 for Display Menu): 1 Enter String: Mayank Length of the String is 6 ENTER CHOICE (7 for Display Menu):

2. For input =2:

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 2
Enter String: Mayank
Enter second String: Kannamwar
Original First String: Kannamwar
Copied String: Kannamwar

ENTER CHOICE (7 for Display Menu): |
```

3. For Input-3:

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 3
Enter String: Mayank
Reversed String: knayaM

ENTER CHOICE (7 for Display Menu):
```

4. For Input = 4 (When Pallindrome=True):

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 4
Enter String: madam
The String is Palindrome.

ENTER CHOICE (7 for Display Menu): |
```

4. For Input = 4 (When Pallindrome=False):

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 4
Enter String: Mayank
The String is NOT Palindrome.

ENTER CHOICE (7 for Display Menu): |
```

5. For Input = 5 (When Substring is present):

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 5
Enter String: String
Enter substring String: Str
The String B is substring of A and occurs 1 time(s).

ENTER CHOICE (7 for Display Menu):
```

5. For Input = 5 (When Substring is not present):

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 5
Enter String: string
Enter substring String: sing
The String B is NOT substring of A.

ENTER CHOICE (7 for Display Menu):
```

6. For Input = 6 (When String are not equal):

```
Output
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 6
Enter String: Mayank
Enter second String: Kannamwar
Strings are NOT Equal.

ENTER CHOICE (7 for Display Menu):
```

6. For Input = 6 (When String are equal):

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 6
Enter String: Mayank
Enter second String: Mayank
Strings are Equal.

ENTER CHOICE (7 for Display Menu): |
```

7. For Input = 7:

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): 7
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

ENTER CHOICE (7 for Display Menu): |
```

8. For Input = 8:

Output

```
<<-----Menu----->>
```

1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display Menu
8. Exit

```
ENTER CHOICE (7 for Display Menu): 8
```

```
=== Code Execution Successful ===
```

Experiment 2:

```
Enter the number of elements:2
Enter 2 elements
5 3
```

```
Array elements are:
5 3
```

```
----MENU----
```

1. Linear Search
2. Binary Search
3. Display Array
4. Exit

```
Enter your choice: 1
```

```
Enter the elements to search: 2
Element is not found
```

```
----MENU----
```

1. Linear Search
2. Binary Search
3. Display Array
4. Exit

```
Enter your choice: 2
```

```
Enter the element to search: 5
Element found at position 1
```

```
----MENU----
```

1. Linear Search
2. Binary Search
3. Display Array
4. Exit

```
Enter your choice: 3
```

```
5 3
```

EXPERIMENT 3:

```

Enter number of elements: 5
Enter elements:
4 5 6 9 8

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 1
Array after Bubble Sort:
4 5 6 8 9

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 2
Array after Selection Sort:
4 5 6 8 9

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 3
Array after Insertion Sort:
4 5 6 8 9

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 4
Array elements are:
4 5 6 8 9

--- SORTING MENU ---
1. Bubble Sort
2. Selection Sort
3. Insertion Sort
4. Display Array
5. Exit
Enter your choice: 5
Exiting program...

...Program finished with exit code 0
Press ENTER to exit console.
```

EXPERIMENT 4:

Choice = 1:

```
| _ _ _ _ _  
  
===== STUDENT MENU =====  
1. Create Student Details  
2. Display Student Details  
3. Linear Search  
4. Sort by Marks  
5. Exit  
  
Enter your choice: 1  
  
Enter details of Student 1  
Enter Roll No: 1  
Enter Name: Radheshyam  
Enter Marks: 7  
  
Enter details of Student 2  
Enter Roll No: onkar  
Enter Name: Enter Marks: 49  
  
Enter details of Student 3  
Enter Roll No: Mayank  
Enter Name: Enter Marks: 28  
  
Enter details of Student 4  
Enter Roll No: 5  
Enter Name: Chunky  
Enter Marks: 69  
  
Enter details of Student 5  
Enter Roll No: 4  
Enter Name: Tanny  
Enter Marks: 0.69
```

Choice = 2 :

```
Enter your choice: 2  
--- Student Details ---
```

```
Student 1  
Roll No: 1  
Name: Radheshyam  
Marks: 7.00
```

```
Student 2  
Roll No: 25771  
Name: onkar  
Marks: 49.00
```

```
Student 3  
Roll No: 1285062377  
Name: Mayank  
Marks: 28.00
```

```
Student 4  
Roll No: 5  
Name: Chunky  
Marks: 69.00
```

```
Student 5  
Roll No: 4  
Name: Tanny  
Marks: 0.69
```

Choice = 3 :

```
Enter your choice: 3  
Enter Roll No to search: 5  
  
Student Found!  
Roll No: 5  
Name: Chunky  
Marks: 69.00
```

Choice = 4 :

```
Enter your choice: 4  
  
Students are sorted according to their marks
```

Choice = 5:

===== STUDENT MENU =====

1. Create Student Details
2. Display Student Details
3. Linear Search
4. Sort by Marks
5. Exit

Enter your choice: 5

Exiting program...

...Program finished with exit code 0

Press ENTER to exit console.

EXPERIMENT 5:

```

    Main Menu
1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 3

Queue is Empty!

Do you want to continue? (y/n): y

    Main Menu
1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 4

Exit...

=== Code Execution Successful ===
```